Network Services

University Teacher: Rahma Haouas Zahwanie

rahma.haouaszahwanie@gmail.com



University Year: 2026 - 2027

PRISTINI
SCHOOL OF AI

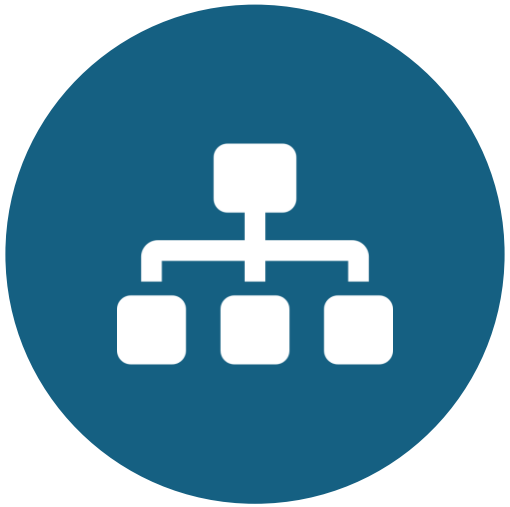
Chapter 2:
Switched Networks and VLANs

Learning Objectives

At the end of this chapter, students will be able to:

- Describe how a company LAN is designed: **converged networks** and the **hierarchical model** (access, distribution, core).
- Explain how a switch **learns, floods, forwards and filters** frames, and compare store-and-forward and cut-through switching.
- Identify **collision domains** and **broadcast domains** in a topology.
- Explain the benefits and the types of **VLANs** (data, default, native, management, voice) and the **802.1Q** trunk.
- Configure and verify VLANs, access ports, voice VLANs and trunks on **Cisco IOS** (and read the Huawei equivalent).
- Apply industry **best practices** and **troubleshoot** common VLAN and trunk problems.

1. Switched Networks

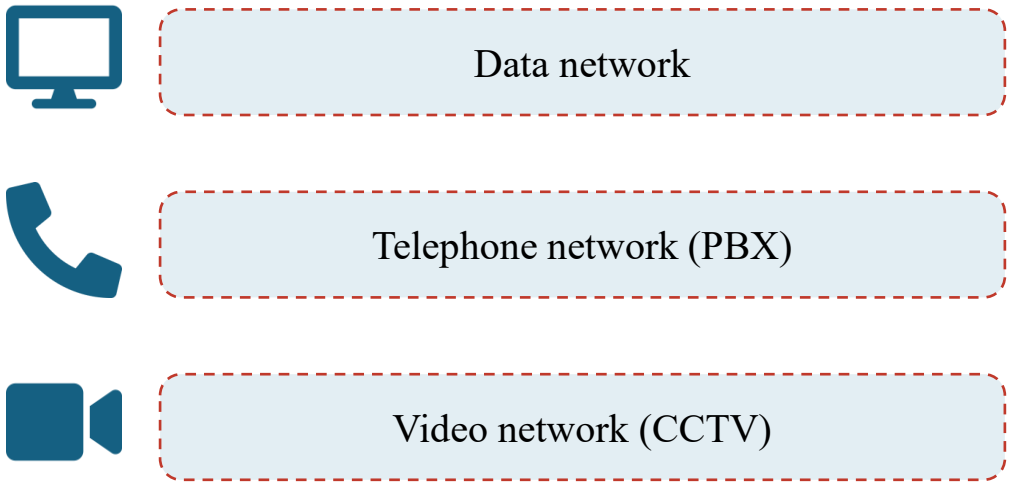


In this part:

- LAN design: converged networks
- The hierarchical model: access, distribution, core
- Real access switches: ports, PoE, uplinks
- How a switch forwards frames (MAC table)
- Collision domains and broadcast domains

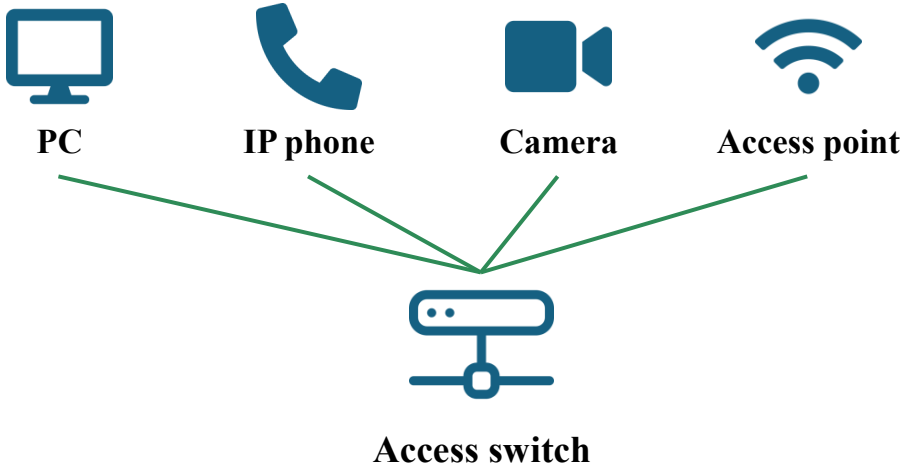
1. LAN Design: Converged Networks

Before: one network per service



3 cablings, 3 teams, 3 budgets.

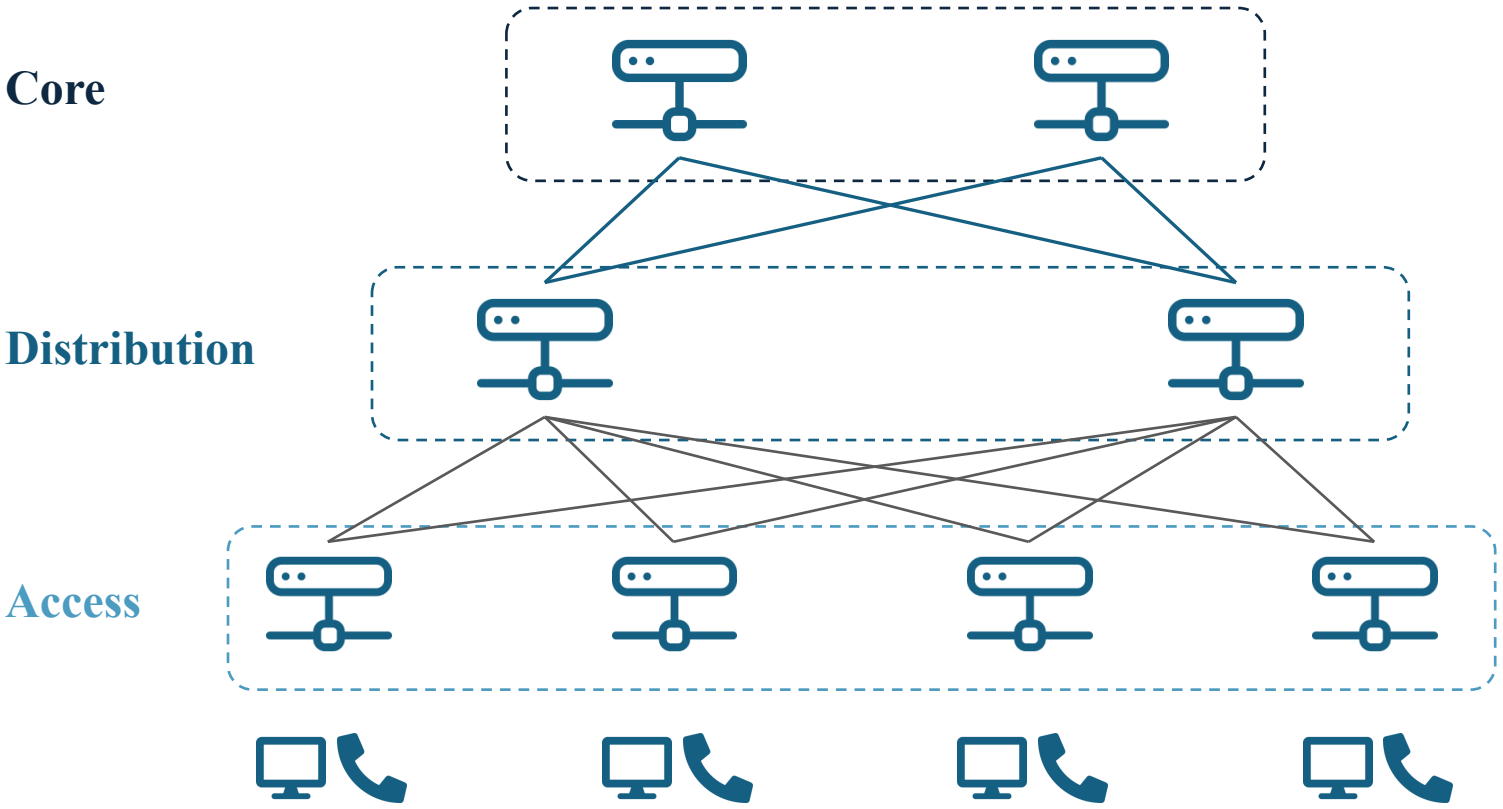
Today: one converged network



- Data, voice and video share **one** cabling and **one** switch.
- Switches give power by cable (**PoE**) and **priority** to voice (QoS).

Converged network = one infrastructure for data, voice and video. It is cheaper, but it must be **well segmented**.

1. LAN Design: The Hierarchical Model



Layer	Role
Core	Very fast transport between buildings; no filtering
Distribution	Joins access switches; routing between VLANs, security rules
Access	Connects end devices; VLANs, PoE, port security

Two links between layers = **redundancy**: if one cable or switch fails, traffic takes the other path.

1. LAN Design: Three-Tier or Collapsed Core?

	Three-tier	Two-tier (collapsed core)
Layers	Core + distribution + access	Core/distribution merged + access
Typical size	Large campus, many buildings	One building, small and medium companies
Cost	Higher	Lower: fewer switches
Growth	Easy to add buildings	Limited
Example	University campus, hospital, big factory	Bank agency, SME, one-floor office

1. LAN Design: What a Real Access Switch Looks Like



24 or 48 RJ45 ports (1 Gb/s), often PoE *Uplinks SFP+ 10 Gb/s*

PoE
Powers phones, Wi-Fi access points and cameras through the network cable.

Managed
Has an IP address, a CLI and a web interface: VLANs, security, monitoring.

Uplinks
Faster ports (10 Gb/s, fiber) towards the distribution switches.

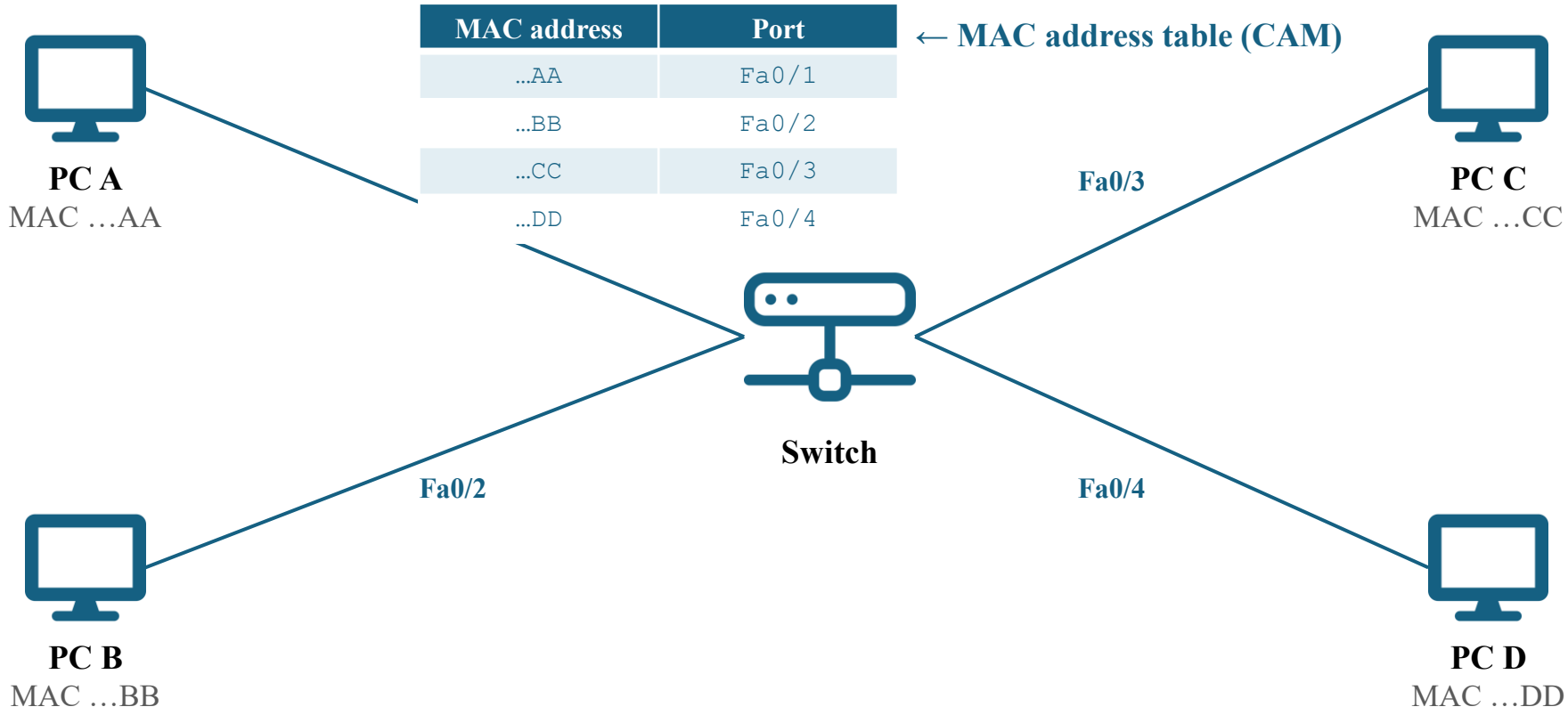
Stacking
Several switches managed as one (one IP, one configuration).

Layer	Example product lines
Access	Cisco Catalyst 9200 / 9300, Huawei S5700
Core / distribution	Cisco Catalyst 9500, Huawei S12700
In the lab	Cisco 2960 (Packet Tracer)

Unmanaged switches (home, small shops) have **no VLANs**: never use them in a company LAN.

Ask for the **datasheet** of a switch: number of ports, PoE budget (W), uplinks, and supported features.

2. Switched Environment: How a Switch Decides



Two questions for each frame: **Where does it come from?** (learn) and **Where does it go?** (forward)

2. Switched Environment: Learn, Flood, Forward, Filter

1. **Learn** — Read the **source** MAC of the frame and write it in the table with the incoming port.
2. **Flood** — Destination **unknown**, or **broadcast** (FF:FF…): send out of **all** ports except the incoming one.
3. **Forward** — Destination **known**: send out of **one** port only.
4. **Filter** — Destination is on the **same** port it came from: drop the frame.

```
S1# show mac address-table
Vlan   Mac Address       Type        Ports
----   -----------       ----        -----
   1   0001.4a2b.11aa    DYNAMIC     Fa0/1
   1   0001.4a2b.22bb    DYNAMIC     Fa0/2
   1   0060.5c3d.33cc    DYNAMIC     Fa0/3
```

Entries are deleted after **300 s** without traffic (aging time, Cisco default).

A switch learns from the **source** address and decides with the **destination** address.

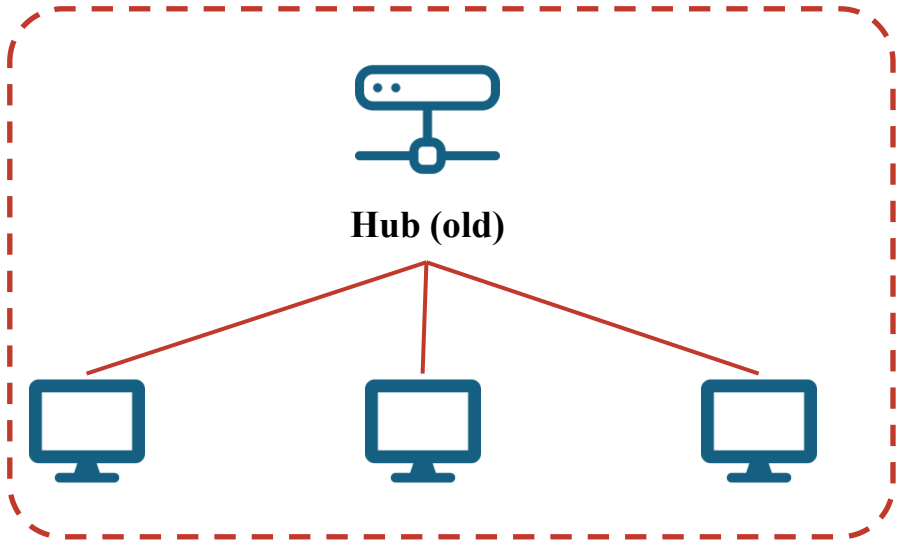
2. Switched Environment: Store-and-Forward or Cut-Through?



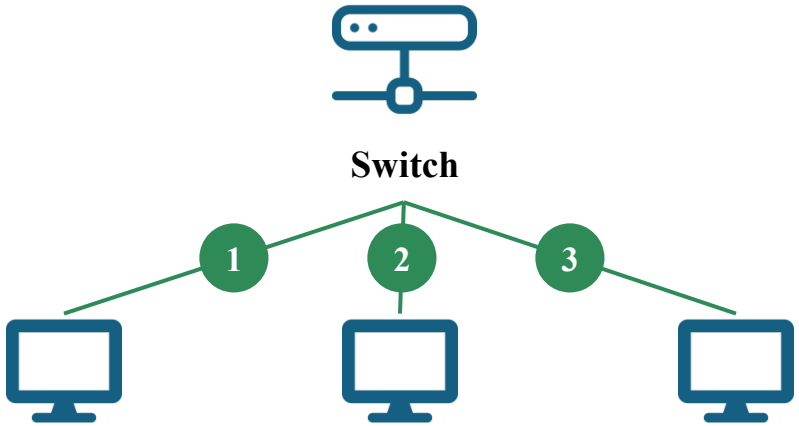
	Store-and-forward	Cut-through
How	Receives the whole frame, checks the FCS (CRC), then sends	Starts sending as soon as it reads the destination MAC
Errors	Damaged frames are dropped	Damaged frames can be forwarded
Speed	A little more delay	Lowest delay (latency)
Used in	Most enterprise switches (default on Cisco Catalyst)	Data centers, high-frequency trading, HPC

*Variant: **fragment-free** = cut-through that waits for the first 64 bytes (where most collision errors are).*

2. Switching Domains: Collision Domains



Hub: **1** collision domain (half duplex)



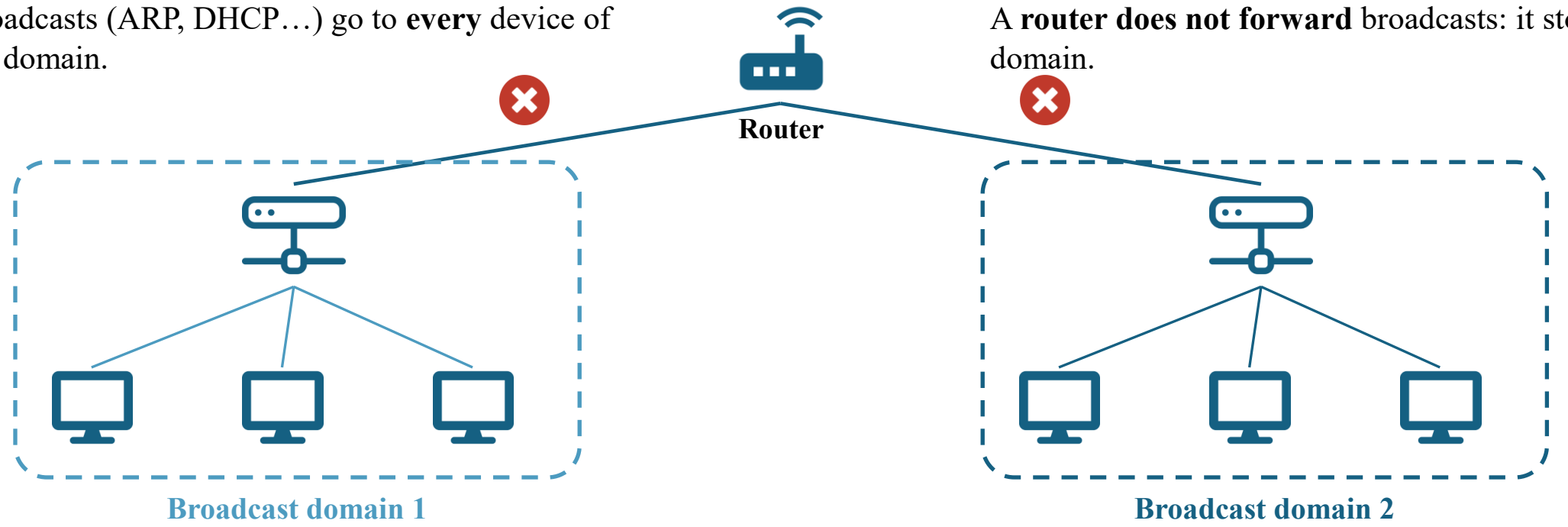
Switch: **1 collision domain per port** (full duplex = no collisions)

Collision domain = the part of the network where two frames sent at the same time collide. Switches split it per port.

2. Switching Domains: Broadcast Domains

Broadcasts (ARP, DHCP…) go to **every** device of the domain.

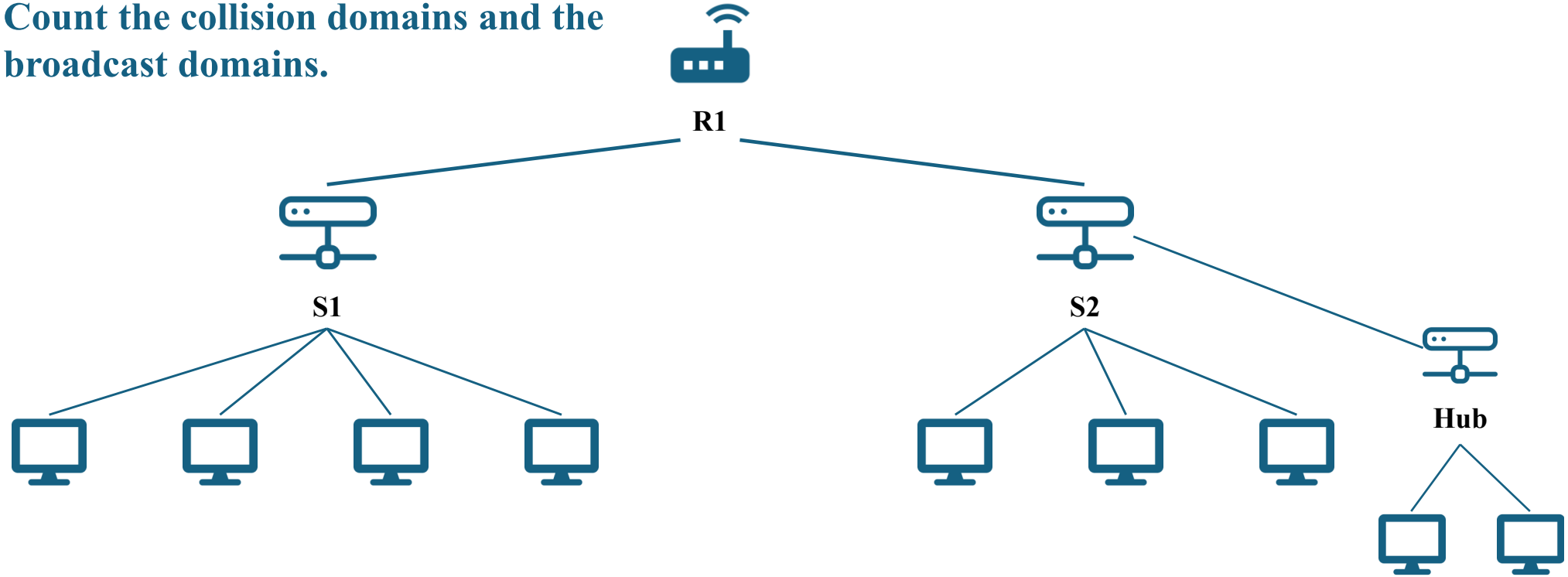
A **router does not forward** broadcasts: it stops the domain.



Switch = 1 broadcast domain for all its ports. Router = 1 broadcast domain **per interface**. **VLAN = 1 broadcast domain** (Part 2).

2. Switching Domains: Exercise 1

Count the collision domains and the broadcast domains.



Hint: each switch port is its own collision domain; the hub shares one. Each router interface ends a broadcast domain.

2. Switching Domains: Solution Exercise 1

Link / segment	Collision domains
S1: 4 PCs + link to R1	5
S2: 3 PCs + link to R1 + link to the hub	5 (the hub and its 2 PCs share one)
Total	10 collision domains

Broadcast domains: 2
one per router interface (S1 side and S2 side). The switches and the hub do not split broadcasts.

Trap: many students count the hub's PCs separately. A hub **repeats** bits: everything behind it is **one** collision domain.

2. Virtual LANs (VLANs)



In this part:

- What a VLAN is and why companies use them
- VLAN types: data, default, native, management, voice
- Trunks and 802.1Q tagging
- Configuration on Cisco IOS (and Huawei)
- Security best practices and troubleshooting
- Beyond the campus: Wi-Fi, data centers, cloud, factories

3. VLAN Segmentation: What Is a VLAN?



- A VLAN (Virtual LAN) groups ports logically, not by place.
- Engineering (VLAN 20) is on 3 floors but is one network.
- Each VLAN = its own broadcast domain and its own IP subnet.
- Devices in different VLANs cannot talk without a router or Layer 3 switch.

1 VLAN = 1 broadcast domain = 1 IP subnet.

3. VLAN Segmentation: Why Companies Use VLANs



Security
Production PLCs and cameras are separated from guests and offices.



Performance
Smaller broadcast domains: less useless traffic on each device.



Lower cost
One switch serves several departments: fewer devices and cables.



Simpler management
Move a user = change one port setting, not the cabling.



Quality of service
Voice gets its own VLAN and priority: no cuts in calls.



Compliance
Banks and hospitals are expected to isolate payment and patient systems.

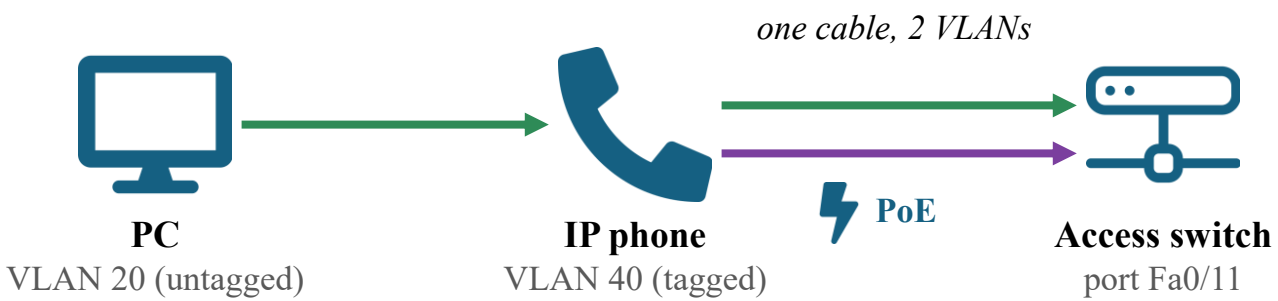
Example: the card payment terminals of a shop should be in an **isolated network** (PCI DSS recommends it). VLANs are the usual first step.

3. VLAN Segmentation: Types of VLANs

Type	What it carries	SahelTech example	Good practice
Data VLAN	Normal user traffic	VLAN 10 ADMIN, 20 ENGINEERING	One per department / function
Default VLAN	All ports at first boot: VLAN 1	/	Do not use VLAN 1 for users
Native VLAN	Untagged traffic on a trunk	VLAN 999 (unused)	Change it from VLAN 1, same on both ends
Management VLAN	Access to the switches (SSH, SNMP)	VLAN 99 MGMT	Separate from users, restricted access
Voice VLAN	IP phone traffic, with priority (QoS)	VLAN 40 VOICE	Separate from data VLAN
Special purpose	OT, cameras, guests	VLAN 30 PRODUCTION, 60 CCTV, 50 GUEST	Strict isolation

In real projects, VLAN **names** are as important as numbers: the next engineer must understand your network.

3. VLAN Segmentation: The Voice VLAN




```
S1(config)# interface fa0/11
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# switchport voice vlan 40
```

- In offices, the PC is often plugged into the phone: only one cable per desk.
- The phone tags its voice frames with VLAN 40 and a priority (CoS 5); the PC's frames stay in VLAN 20.
- The switch learns the voice VLAN of the phone automatically (CDP or LLDP).

Result: calls stay clear even when the PC downloads a big file.

3. VLAN Segmentation: VLAN ID Ranges



	Normal range	Extended range
IDs	1 - 1005 (1002-1005 reserved for old Token Ring / FDDI)	1006 - 4094
Stored in (Cisco)	vlan.dat in flash memory	running configuration
Typical use	Enterprise and campus networks	Service providers, large data centers

Why 4094? The VLAN ID field in the 802.1Q tag has **12 bits**: $2^{12} = 4096$ values, minus 0 and 4095.

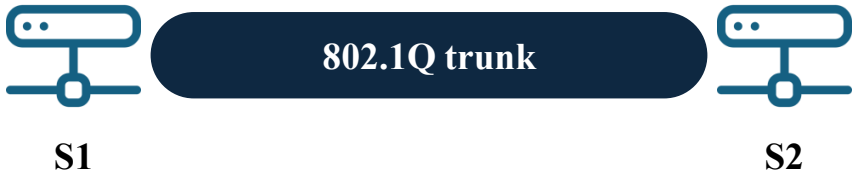
4. VLANs Across Switches: Why We Need Trunks

Without trunk: one cable per VLAN



- 4 VLANs = 4 cables and 8 ports.
- Does not scale: 20 VLANs = 20 cables!

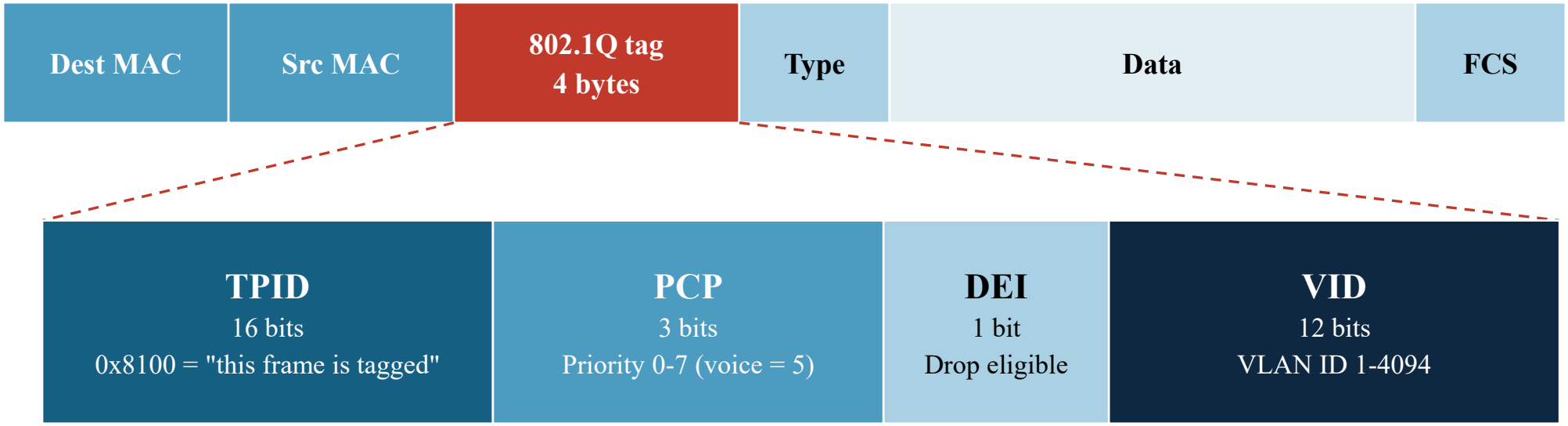
With a trunk: one link for all VLANs



- Each frame gets a **tag** with its VLAN number.
- Used between switches, and towards routers, servers and Wi-Fi APs.

Port type	Carries	Typical connection
Access port	One VLAN, untagged frames	PC, printer, camera, PLC
Trunk port	Many VLANs, tagged frames (802.1Q)	Switch ↔ switch, switch ↔ router / AP / server

4. VLANs Across Switches: The 802.1Q Tag



- The tag is added when a frame enters a trunk and removed before it leaves on an access port: PCs never see it.
- The switch recalculates the FCS because the frame changed.

Wireshark tip: a tagged frame shows a line **802.1Q Virtual LAN, PRI: 5, ID: 40** between Ethernet and IP.

4. VLANs Across Switches: A Frame Crosses the Trunk

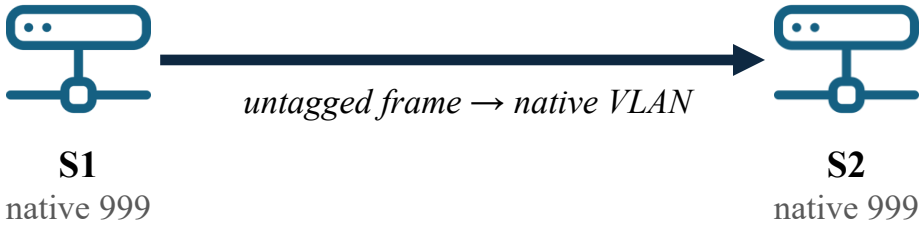


1. PC1 sends a normal Ethernet frame (no tag) on its access port.
2. S1 knows the port is in VLAN 20; the destination is behind the trunk → it **adds tag 20**.
3. S2 reads tag 20, finds PC2 in VLAN 20 → it **removes the tag** and forwards.

If VLAN 20 is **not allowed** on the trunk, or **does not exist** on S2, the frame is dropped: PC1 and PC2 cannot talk. (Most common lab error)

Only the **trunk** carries tags. Access ports and end devices work with **normal frames**.

4. VLANs Across Switches: The Native VLAN



Log message

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN
mismatch discovered on GigabitEthernet0/1
(999), with S2 GigabitEthernet0/1 (1).
```

- On a trunk, frames without tag belong to the native VLAN.
- Default native VLAN = VLAN 1.
- Both ends must use the same native VLAN, or traffic leaks between VLANs.

Best practice

- Use an unused VLAN as native (e.g. 999), with no port in it.
- This blocks the double-tagging attack.

Native VLAN mismatch = classic exam question and classic real-life incident after replacing a switch.

5. VLAN Implementation: The Configuration Workflow



Prompt	Mode	How to enter
`S1>`	User EXEC	`login`
`S1#`	Privileged EXEC	`enable`
`S1(config)#`	Global configuration	`configure terminal`
`S1(config-if)#`	Interface configuration	`interface fa0/1`

5. VLAN Implementation: Creating VLANs

```
S1# configure terminal
S1(config)# vlan 10
S1(config-vlan)# name ADMIN
S1(config-vlan)# vlan 20
S1(config-vlan)# name ENGINEERING
S1(config-vlan)# vlan 40
S1(config-vlan)# name VOICE
S1(config-vlan)# vlan 99
S1(config-vlan)# name MGMT
S1(config-vlan)# vlan 999
S1(config-vlan)# name NATIVE_UNUSED
S1(config-vlan)# end
```

- `vlan 10` creates the VLAN (or enters it if it exists).
- `name` is optional but always used in companies.
- The VLAN must exist on every switch where it is used (or be learned with VTP, not used in this course).
- `no vlan 10` deletes it: its ports become inactive

Naming convention example: UPPERCASE, function first: ADMIN, ENGINEERING, VOICE, CCTV, GUEST, MGMT.

5. VLAN Implementation: Assigning Access Ports

```
S1(config)# interface range fa0/1 - 10
S1(config-if-range)# description ADMIN users
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 10
S1(config-if-range)# exit
S1(config)# interface range fa0/11 - 20
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 20
S1(config-if-range)# switchport voice vlan 40
```

- `interface range` configures many ports at once.
- `switchport mode access` fixes the port as access (no negotiation).
- `description` helps the support team: **always** write it.
- Ports 11-20: PC in VLAN 20 + phone in VLAN 40.

A port belongs to **VLAN 1** until you assign it. Unused ports: put them in a "parking" VLAN and **shut them down**.

5. VLAN Implementation: Configuring a Trunk

```
S1(config)# interface g0/1
S1(config-if)# description TRUNK to S2
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 999
S1(config-if)# switchport trunk allowed vlan 10,20,40,99
S1(config-if)# switchport nonegotiate
```

Command	Why
`mode trunk`	Force trunk
`native vlan 999`	Safe native VLAN
`allowed vlan`	Only needed VLANs
`nonegotiate`	Disable DTP

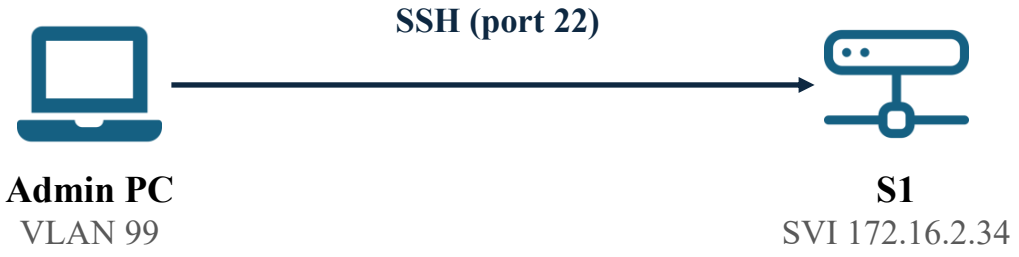
Careful: `switchport trunk allowed vlan 30` replaces the list. To add a VLAN, use `switchport trunk allowed vlan add 30`.

On some switches (e.g. Cisco 3560) you first type `switchport trunk encapsulation dot1q`. The 2960 used in Packet Tracer supports only 802.1Q.

5. VLAN Implementation: Remote Management (SVI)

```
S1(config)# interface vlan 99
S1(config-if)# ip address 172.16.2.34 255.255.255.240
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 172.16.2.33
```

- A Layer 2 switch has **no IP** on its ports.
- The **SVI** (Switch Virtual Interface) gives it an IP in the management VLAN.
- `ip default-gateway` lets admins reach it from another subnet.



In industry, admins manage hundreds of switches by **SSH**, never Telnet, and only from the management VLAN.

Management VLAN = a **separate door** for administrators. Users must not see it.

5. VLAN Implementation: Verify: show vlan brief

```
S1# show vlan brief
VLAN Name                Status    Ports
---- ----------------- --------- ----------------------
1    default             active    Fa0/21, Fa0/22, Fa0/23
10   ADMIN               active    Fa0/1, Fa0/2, ... Fa0/10
20   ENGINEERING         active    Fa0/11, Fa0/12, ... Fa0/20
40   VOICE               active    Fa0/11, Fa0/12, ... Fa0/20
99   MGMT                active
999  NATIVE_UNUSED       active
```

- Shows **access** ports only: trunk ports are **not** listed.
- Ports 11-20 appear in VLAN 20 **and** 40 (voice).
- Fa0/21-23 still in VLAN 1: forgotten ports

Other useful commands	show interfaces fa0/1 switchport	show interfaces trunk	show mac address-table vlan 20	show running-config

5. VLAN Implementation: show interfaces trunk

```
S1# show interfaces trunk
Port   Mode  Encapsulation  Status    Native vlan
Gi0/1  on    802.1q         trunking  999

Port   Vlans allowed on trunk
Gi0/1  10,20,40,99

Port   Vlans allowed and active in management domain
Gi0/1  10,20,40,99
```

- **Status trunking** = the trunk works.
- **Native vlan** must be identical on both ends.
- **Vlans allowed**: is your VLAN in the list?
- Run it on **both** switches and compare.

Real support work = reading these outputs fast. Practice until you can find an error in **30 seconds**.

5. VLAN Implementation: Same Job, Another Vendor

Task	Cisco IOS	Huawei VRP
Create VLAN	`vlan 10`	`vlan 10`
Access port	`switchport mode access`	`port link-type access`
Put port in VLAN	`switchport access vlan 10`	`port default vlan 10`
Trunk port	`switchport mode trunk`	`port link-type trunk`
Allowed VLANs	`switchport trunk allowed vlan 10,20`	`port trunk allow-pass vlan 10 20`
Show VLANs	`show vlan brief`	`display vlan`
Save	`copy running-config startup-config`	`save`

Huawei equipment is common in Tunisian operators and companies. Same concepts, different words: `show` ↔ `display`.

5. VLAN Implementation: DTP: Why We Turn It Off

S1 mode \ S2 mode	dynamic auto	dynamic desirable	trunk	access
dynamic auto	Access	Trunk	Trunk	Access
dynamic desirable	Trunk	Trunk	Trunk	Access
trunk	Trunk	Trunk	Trunk	Limited
access	Access	Access	Limited	Access

- **DTP** (Dynamic Trunking Protocol, Cisco) lets ports **negotiate** trunk or access.
- Convenient, but an attacker's PC can pretend to be a switch and get a trunk.
- Industry rule: fix every port: `mode access` or `mode trunk` + `nonegotiate`.

Never leave a port to chance: decide its mode yourself.

6. VLAN Security: VLAN Hopping Attacks

1. Switch spoofing

The attacker's PC speaks DTP and becomes a trunk: it receives traffic of all VLANs.

Protection: `switchport mode access` + `switchport nonegotiate` on user ports.

2. Double tagging

The attacker, in the native VLAN, sends a frame with two tags. The first switch removes the outer one; the next switch sends it into the victim VLAN.

Protection: native VLAN = **unused** VLAN, no users in it.

VLANs separate traffic, but they are not a firewall. Sensitive zones (OT, payments) also need filtering rules.

6. VLAN Security: Industry Best Practices Checklist

- No users in **VLAN 1**
- Separate **management** VLAN, SSH only
- **Disable DTP** (`nonegotiate`)
- **Description** on every port and VLAN name
- Native VLAN = **unused** VLAN (e.g. 999)
- Unused ports: **shutdown** + parking VLAN
- Trunks: **allow only** the needed VLANs
- Document the VLAN plan (IDs, names, subnets)

Network audits (and job interviews!) check exactly these points.

7. Troubleshooting: Common VLAN Problems

Symptom	Likely cause	Check with	Fix
PC has no network, port inactive	VLAN deleted or not created	`show vlan brief`	Create the VLAN again
Same VLAN, different switches: no ping	VLAN not allowed on trunk	`show interfaces trunk`	allowed vlan add X
Log: NATIVE_VLAN_MISMATCH	Native VLAN differs on the two ends	`show interfaces trunk`	Same native VLAN
Link up but no trunk	Both ends dynamic auto, or access/trunk mismatch	`show interfaces g0/1 switchport`	mode trunk on both
PC in right VLAN, no ping	Wrong IP subnet for this VLAN	`ipconfig / ip addr`	Correct IP / mask

Work **bottom-up**: cable/port up? → right VLAN? → trunk carries it? → right IP subnet?

Exercise 2: MAC Address Table

Switch S1 just booted (empty table). PC A is on Fa0/1, PC B on Fa0/2, PC C on Fa0/3. For each frame, say what S1 does (learn? flood or forward?) and write the table after each step.

Step	Frame	S1 action	MAC table after
1	A → B	?	?
2	B → A	?	?
3	C → FF:FF:FF:FF:FF:FF (ARP)	?	?
4	A → C	?	?

Exercise 2: Solution

Step	Frame	S1 action	MAC table after
1	A → B	Learn A on Fa0/1; B unknown → **flood** (Fa0/2, Fa0/3)	A-Fa0/1
2	B → A	Learn B on Fa0/2; A known → **forward** to Fa0/1	A-Fa0/1, B-Fa0/2
3	C → broadcast	Learn C on Fa0/3; broadcast → **flood** (Fa0/1, Fa0/2)	A, B, C
4	A → C	A already known; C known → **forward** to Fa0/3	A, B, C (A's timer reset)

Broadcasts are **always** flooded, even when the table is full. That's why big flat networks suffer.

Exercise 3: Design the SahelTech VLAN Plan

Use **172.16.0.0/16** and **VLSM**. Give each VLAN an ID, a name, a subnet and a gateway (first usable address). Start with the biggest need.

Group	Hosts needed
Guest Wi-Fi	200
IP phones	120
Administration	50
Engineering	28
Cameras (CCTV)	25
Production (PLCs)	20
Switch management	10

Also decide:

- the native VLAN ID
- which VLANs go on the trunk between the two buildings
- which VLANs must **never** reach the guests

Exercise 3: Solution (one possible plan)

VLAN	Name	Hosts	Subnet	Usable	Gateway
50	GUEST	200	172.16.0.0/24	254	172.16.0.1
40	VOICE	120	172.16.1.0/25	126	172.16.1.1
10	ADMIN	50	172.16.1.128/26	62	172.16.1.129
20	ENGINEERING	28	172.16.1.192/27	30	172.16.1.193
60	CCTV	25	172.16.1.224/27	30	172.16.1.225
30	PRODUCTION	20	172.16.2.0/27	30	172.16.2.1
99	MGMT	10	172.16.2.32/28	14	172.16.2.33
999	NATIVE_UNUSED	0	—	—	—

Trunk between buildings: VLANs **10, 20, 30, 40, 60, 99** (guests only where there are access points). Guests must never reach VLAN 30 (PLCs): this is enforced by router/firewall rules in Chapter 3.

In many real companies, admins simply give a /24 per VLAN in 10.x.x.x: easier to read, plenty of addresses.

Exercise 4: Configure an Access Switch



Write the full configuration of **SW-F1** (first floor, Cisco 2960) with the plan of Exercise 3:

1. Create VLANs 10, 20, 40, 99 and 999 with their names.
2. Ports Fa0/1–10: ADMIN. Ports Fa0/11–20: ENGINEERING with IP phones (VOICE).
3. Ports Fa0/21–24: not used → shut down.
4. Gi0/1: trunk to the distribution switch, native VLAN 999, only VLANs 10, 20, 40, 99, no DTP.
5. Management IP: **172.16.2.35/28**, gateway 172.16.2.33.
6. Which two commands prove that your configuration works?

Exercise 4: Solution

```
SW-F1(config)# vlan 10
SW-F1(config-vlan)# name ADMIN
! same for 20 ENGINEERING, 40 VOICE,
! 99 MGMT, 999 NATIVE_UNUSED
SW-F1(config)# interface range fa0/1 - 10
SW-F1(config-if-range)# switchport mode access
SW-F1(config-if-range)# switchport access vlan 10
SW-F1(config)# interface range fa0/11 - 20
SW-F1(config-if-range)# switchport mode access
SW-F1(config-if-range)# switchport access vlan 20
SW-F1(config-if-range)# switchport voice vlan 40
SW-F1(config)# interface range fa0/21 - 24
SW-F1(config-if-range)# shutdown
```

```
SW-F1(config)# interface g0/1
SW-F1(config-if)# switchport mode trunk
SW-F1(config-if)# switchport trunk native vlan 999
SW-F1(config-if)# switchport trunk allowed
                  vlan 10,20,40,99
SW-F1(config-if)# switchport nonegotiate
SW-F1(config)# interface vlan 99
SW-F1(config-if)# ip address 172.16.2.35
                  255.255.255.240
SW-F1(config-if)# no shutdown
SW-F1(config)# ip default-gateway 172.16.2.33
```

Proof: `show vlan brief` (ports in the right VLANs) and `show interfaces trunk` (native 999, allowed list). Then `copy run start`.

Exercise 5: Troubleshooting



PC1 (VLAN 20, on S1) cannot ping PC2 (VLAN 20, on S2). The IP addresses are correct. Find **two** problems and give the fix.

S1

```
S1# show interfaces trunk
Port    Mode  Encapsulation  Status    Native vlan
Gi0/1   on    802.1q         trunking  999
Port    Vlans allowed on trunk
Gi0/1   10,40,99
```

S2

```
S2# show interfaces trunk
Port    Mode  Status    Native vlan
Gi0/1   on    trunking  1
Port    Vlans allowed on trunk
Gi0/1   1-4094
```

Hint: compare the two outputs line by line, like a support engineer.

Exercise 5: Solution

#	Problem	Evidence	Fix
1	VLAN 20 not allowed on S1's trunk	S1 allowed list: 10,40,99	`S1(config-if)# switchport trunk allowed vlan add 20`
2	Native VLAN mismatch (999 vs 1)	Native vlan column differs; CDP log message	`S2(config-if)# switchport trunk native vlan 999`

Also good practice on S2: restrict its allowed list to the needed VLANs instead of 1-4094.

Notice `add`: without it, the trunk would carry **only** VLAN 20 and break VLANs 10, 40 and 99!

In industry

8. Beyond the Campus: VLANs in Industry

Wi-Fi

Each SSID is mapped to a VLAN: "SahelTech-Staff" → VLAN 10, "SahelTech-Guest" → VLAN 50. The AP is connected by a trunk.

Factories (OT)

IT and OT (machines, PLCs) are separated in zones. Standard IEC 62443 calls them **zones and conduits**. VLANs are the first layer.

Data centers

4094 VLANs are not enough for big data centers: they use **VXLAN** (24-bit ID ≈ 16 million segments) on top of IP.

Cloud

In AWS or Azure you do not configure VLANs: you create **VPCs / VNets and subnets**. Same idea: isolate and segment.

What you learn today on a Cisco 2960 is the **same idea** you will use in factories, data centers and the cloud.

8. Beyond the Campus: Tools of the Profession

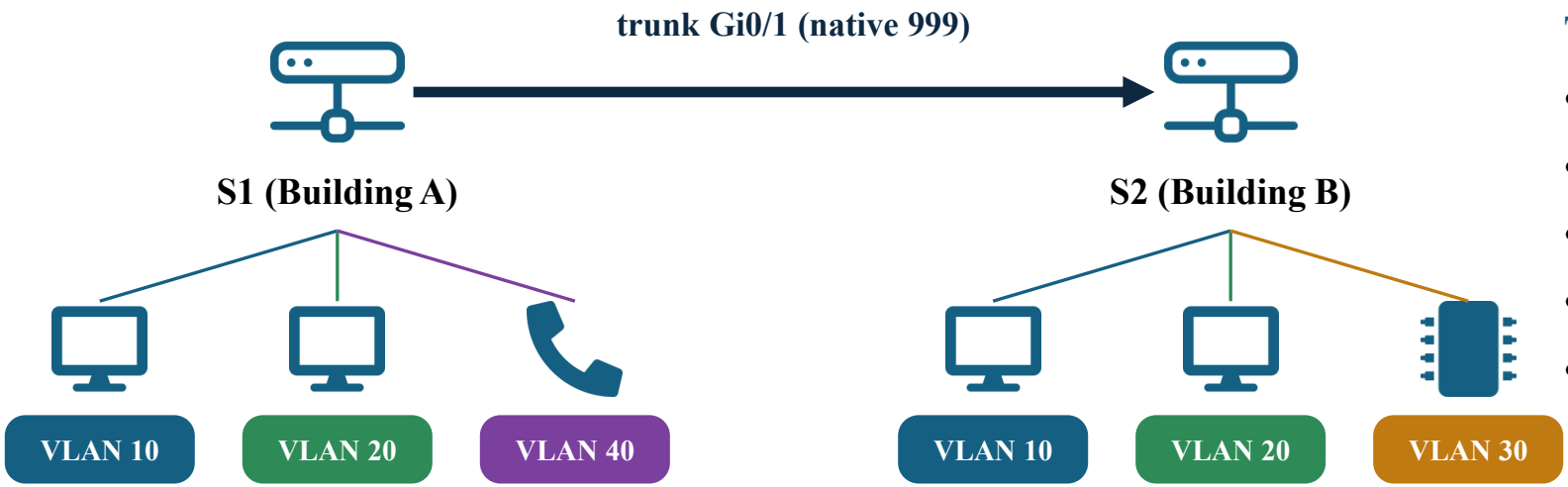
Tool	Used for
Cisco Packet Tracer	Learning and CCNA labs (free with Cisco NetAcad)
GNS3 / EVE-NG / Cisco CML	Real device images, used by professionals to test before production
Wireshark	See 802.1Q tags, ARP and broadcasts
Network diagrams (draw.io, Visio)	Document the VLAN plan
Ansible / Python (Netmiko)	Configure hundreds of switches automatically

Your next steps

- Finish the Packet Tracer labs of this module.
- Follow the Cisco NetAcad course **CCNA: Switching, Routing and Wireless Essentials**.
- Put your lab topologies on **GitHub** with their configs: recruiters look at it.
- Target the **CCNA 200-301** certification.

Network automation (Python, Ansible) is a **growing job market**: software engineers who understand VLANs are rare.

Lab TP1: Preview



Tasks

- Create VLANs and names
- Assign access + voice ports
- Configure and secure the trunk
- Management SVI + SSH
- Test: ping inside / between VLANs

Question to answer in the report: why can a PC in VLAN 10 ping another PC in VLAN 10 on the other switch, but **not** a PC in VLAN 20? → answer in **Chapter 3** (inter-VLAN routing).

Deliverable: the .pkt file + `show vlan brief` and `show interfaces trunk` of both switches.

Cheat Sheet: Switching and VLAN Commands

Create & assign

`vlan 10 / name ADMIN`	create a VLAN
`interface range fa0/1 - 10`	many ports
`switchport mode access`	fix access mode
`switchport access vlan 10`	port in VLAN 10
`switchport voice vlan 40`	IP phone VLAN
`shutdown`	unused ports

Trunks

`switchport mode trunk`	fix trunk mode
`switchport trunk native vlan 999`	safe native VLAN
`switchport trunk allowed vlan …`	set the list
`… allowed vlan add 30`	add to the list
`switchport nonegotiate`	disable DTP
`interface vlan 99 + ip address`	management SVI

Verify

`show vlan brief`	VLANs and access ports
`show interfaces trunk`	trunks, native, allowed
`show interfaces fa0/1 switchport`	one port in detail
`show mac address-table`	learned MACs
`show running-config`	full configuration
`copy run start`	save

Huawei equivalents

`vlan 10`	create a VLAN
`port link-type access`	access mode
`port default vlan 10`	port in VLAN 10
`port link-type trunk`	trunk mode
`port trunk allow-pass vlan 10 20`	allowed VLANs
`display vlan`	verify

Summary

In this chapter, you learned to:

- Design a company LAN with converged services and the hierarchical model (access, distribution, core).
- Explain how a switch learns, floods, forwards and filters, and count collision and broadcast domains.
- Use VLANs to split broadcast domains: 1 VLAN = 1 broadcast domain = 1 IP subnet.
- Carry many VLANs on one 802.1Q trunk, with a safe native VLAN.
- Configure, verify and troubleshoot VLANs on Cisco IOS, and recognize the Huawei syntax.
- Apply the security checklist used in industry.



THE END